Requirement Analysis

Weighting: 12% | **5 pages**

- Deadline: 16 February 2026
- **Marking Guidelines (https://canvas.liverpool.ac.uk/courses/86513/pages/assessment)**
- Team Submission
- **Submission Details (https://canvas.liverpool.ac.uk/courses/86513/assignments/332737)**

The first few weeks of term, once the team is formed, should be spent defining the chosen project requirements. Therefore, the first review exercise looks at your team's project proposal and the list of requirements contained in it.

The main aim of the exercise is to stipulate a ***contract***, provide each team with a reasonable project idea that has been checked, validated, and sometimes slightly modified by the module teaching team, to fulfil the course aims. A side effect of this is an assessment of your requirement analysis efforts. From the end of the assessment exercise, your commented proposal summary should be used as a constant reference for what needs to be done.

Proposal Summary

Each team is required to submit a 5-page proposal summary. The purpose of the proposal summary is to show that you have a clear idea of what the project involves, and a well-defined plan for realising it. The document should be concise but clear and specific, following the structure outlined below. The summary must fit in five pages, all included. Documents deviating from this limit will attract lower grades.

Key Questions

Writing the proposal helps both you and us to understand what the project is all about and what the end product will be. At the simplest level, it needs to answer three broad questions.

- What will you do?
- How will you do it?
- What will it look like?

If your project involves designing and implementing a piece of software (as most projects do), then the proposal should clearly answer these questions. A proposal with more of a research focus will still cover these things, but you would describe your experimental process instead. The key thing is that the document should provide **sufficient detail** using **non-specialist language**, so anyone with a general computer science background would know what your project involves.

CLARITY IS KEY

The information may be there, but it needs to be presented properly.

Title

Your project should have a title. A concise textual expression providing a quick description of what the project is about.

Project Description

This should be a high-level overview of your project, suitable for a reader who isn't an expert in the topic area. This will be most useful for your marker, who might have no prior knowledge of the project. This first section should be relatively short and succinct, but make sure you answer the key question of what you will be doing on the project. You can add more detail in the later sections.

Aims & Objectives

The aims of your project are broad statements of intent. In other words, what you expect to have achieved after completing the work. For example, *Create a platform that allows users to design their own board games*. Objectives are more measurable and include the steps needed to achieve the aims. For example, *Implement a game description language that encodes the rules for board games*.

There will usually be more objectives than aims. The difference between aims and objectives is not always easy to grasp, but objectives should be things that you can test. In the given example, the testing plan would ensure the game description language is robust and works properly. Your aims and objectives should be presented as bullet points to help readability.

Key Literature & Background Reading

During the first couple of weeks of term, you will be conducting background reading and looking at various sources of information. Some will be technical (such as API documentation), and some will cover previous research in the area. In this section, you should demonstrate a good knowledge of the topic area, discussing, citing and summarising key literature, and using it to justify your design and development decisions.

Development & Implementation Summary

Bearing in mind that you won't have fully started development when you write the proposal, this section should summarise your (technical) plans. You should provide some details of your planned design (what will be the system's main components? how will they be linked to each other? how will they relate to the aims and objectives?). Provide a brief overview of your proposed development environment and implementation language, and the reasons why you chose them. Each project is different, so you might have little or lots to explain here. Also, make sure this section covers the key question of how you will implement (realise) your project, including how your workflow will be organised.

Addendum. The main goal here is to convince us that you have put together a work plan that will fill the rest of the term. It is amazing what can be achieved these days with frameworks and a couple of library calls. That will NOT be enough on its own to earn A+ grades. Robust designs, research-level technical solutions, might. If technology is there to save programming efforts, the requirements should be extended to fill the available time.

Data Sources

This is one of the sections in which teams tend to underperform. Avoid as much as possible statements that could apply to any project. We really want to learn about data in your system.

If you are using datasets from the public domain, accessing or processing publicly available text, or generating your own data as part of the project, mention it in this section. You should describe the data you will be using, state how it will be obtained, confirm you are using it with permission, and explain how you will ensure the confidentiality of any personal information. If you're not using any data in your project, then include a statement to that effect, but check with your project monitor that there is genuinely no data use. Remember that evaluation questionnaires will generate data.

Testing & Evaluation

Explain how you will test and evaluate your final product. At this stage, you don't have to include a detailed testing plan, since you probably don't know exactly what components will be produced or how they will work together. However, you should at least have a high-level idea of the type of testing you will carry out. If you intend to use other people as beta testers, including the use of evaluation questionnaires, you should confirm that this will be done ethically (see the next section).

Ethical Considerations

In this section, you should mention anything that might require you to act ethically. As mentioned before, when talking about data, try to avoid generic statements. Focus on your project. You should state that you have read our **ethical guidelines (https://canvas.liverpool.ac.uk/courses/86513/pages/ethics)** and will follow them. This will include the use of data in the public domain, the generation of new data, and the results of testing and evaluation. There could be other ethical considerations that are specific to your project. If so, discuss them with your project monitor and explain them briefly here.

BCS Project Criteria

Your degree programme is accredited by **BCS (https://www.bcs.org/)**, the Chartered Institute for IT. They require projects to demonstrate the following six outcomes.

- An ability to apply practical and analytical skills gained during the degree programme.
- Innovation and/or creativity.
- Synthesis of information, ideas and practices to provide a quality solution together with an evaluation of that solution.
- That your project meets a real need in a wider context.

- An ability to self-manage a significant piece of work.
- Critical self-evaluation of the process.

In this section, you should explain how ***your*** project deliverables and development method will meet these criteria. Some of these might be covered in other sections of the document, so refer to them if necessary. Be specific!! The last thing we want is to read 69 BCS Criteria sections that look exactly the same.

UI/UX Mockup

At this point in the project, you probably won't have a full design for whatever you are developing, but you will have a rough idea of what the UI (user interface) and UX (user experience) will be. This section should answer the key question of what your project will look like. Include a mockup of the user interface, either as a wireframe drawing or a pencil sketch that you scanned in. It doesn't have to be perfect, complete, or show every possible user interaction, and you might change the design later. It just needs to be enough to show that you have thought about what the software will look like and how it will be used.

Project Plan including Risks & Contingency Plans

Planning (https://canvas.liverpool.ac.uk/courses/86513/pages/planning) is crucial for a project run by a group of people, and bad planning is the most frequent cause of problems in a group project.

Include a Gantt chart (or similar project plan) with enough detail to show each stage of your project and your estimates for their duration. Show aspects that can overlap, activities that have dependencies, and so on. Remember to include time for presenting the project and writing all required documentation.

Include a table with four columns: Risks, Contingencies, Likelihood and Impact. Common risks for all projects include hardware failure, software failure, running out of time, and programming problems. List possible risks in the first column, and your plans for preventing or handling them in the second column. The third column should state how likely you think the risk is (low, medium or high), and the final column should state the impact. Sometimes things with very low likelihood, such as losing your backups, could have a very high impact on the project.

References

List references to articles, websites, videos, datasets, journals, and any other background reading you carried out while writing the document and planning your project. References should be cited properly in the main body of the document. You should use the **Harvard** (https://en.wikipedia.org/wiki/Parenthetical_referencing) or **IEEE** (https://en.wikipedia.org/wiki/IEEE_style) referencing style. Guidance is available on the **university library website** (https://libguides.liverpool.ac.uk/referencing/).

Frequently Asked Questions

Q1. Project Scope.

- Can we only include some basic requirements (which we are sure we can implement) in the requirement document, then implement more functions than that?

Answer: Your goal is to be realistic (that's part of good planning). You must submit a proposal that will keep your team (all people in there) busy for three months! And if your team is relatively small, remember that the effect of team size on productivity is inversely proportional to the size of the team. There is plenty of research on this (for instance, check out this YouTube video on the **Ringelmann Effect** (https://www.youtube.com/watch?v=RdcNkg4V594), things can actually get worse the more people are around!)
If in doubt, my advice is to go slightly bold. Prioritise among objectives. Decide which ones are the most likely to be given up if time runs short. Better to cut a little bit (later, if necessary), than to put forward a proposal that is too simple.

Q2. Requirement documents

- Are there any good examples of such a requirement document?

Answer: This is a very tricky question indeed, for me ... and for anybody else (I dare you to find someone on the internet that would give you the answer you are after). Here's few points for you to reflect on:

- While, in simple programming or mathematical tasks, there is often only a single answer, here there are many. Ask a music teacher to define a good interpretation: it's easy to spot a technical mistake ... but I've heard many performances that were not communicating much! The notion of **good** is also influenced by academic judgment, which is a personal thing. I am sure that if one of my colleagues teaching COMP208 and I had to assess the same requirement document .. we would probably **not** come up with the same marks. Each of us might focus on different aspects. But I am almost sure we would recognise the same problematic issues ...
- In any case, the **good** principles are on this page. Connolly and Begg (C&B) give one good example (let me know if you find any other anywhere else!), ... but, please, please, read on ...
- I have been teaching this module for a long time, and I deliberately decided NOT to give "good" examples 'cause then what happens is that if I did, the vast majority of the class would write a detailed proposal fitting it to the "good" example, which is a huge mistake. Techniques must be used to one's advantage, not to trap our ideas in meaningless diagrams and lists. I give u an example. I just said that you could consider the example on C&B as a good source of inspiration ... but now suppose that you want to develop a multi-player network game, or a plant production control system ... probably very little of what is on C&B will be relevant, or "good" for you.

Aim for clarity. Aim for completeness. Aim to show us that you have good plans in place.